

DECK 04 · WEEK 2 · MORNING 2

Microeconomics & Business Cycles

How firms set price and output, and how the economy cycles around its potential path

CFA Level I · Vol 2 Learning Modules 1–2

Roadmap — Microeconomics & Business Cycles

6 sections · ~2 hours · Built bottom-up from CFA readings

01

Demand, supply, elasticity

Law of demand · own-price / cross-price / income elasticity · revenue reaction

02

Firm costs & profit maximization

Economic vs accounting profit · $MR = MC$ · breakeven / shutdown · economies of scale

03

Four market structures

Perfect / monopolistic / oligopoly / monopoly · kinked demand · Cournot · Nash · HHI

04

Business cycle phases & indicators

Four phases · classical vs growth · credit cycles · LEI / coincident / lagging

05

Unemployment & inflation

Labor-force measures · CPI · core · PPI · GDP deflator · Phillips curve · Okun's law

06

Investment-relevant cycle dynamics

Leading markets · inventory and capex cycles · sector rotation playbook

Learning objectives

By the end of this deck you should be able to:

1. Compute and interpret own-price elasticity of demand, and predict how total revenue will move when a firm changes price given the elasticity regime.
2. Apply the $MR = MC$ profit-maximization rule to any market structure, and distinguish the breakeven, shutdown, and long-run-exit thresholds.
3. Identify which of the four market structures a firm operates in using the five determinants, and compute CR_N and HHI.
4. Describe the four business-cycle phases and translate inventory, capex, employment, and credit indicators into a phase call.
5. Link unemployment and inflation through the short-run Phillips curve and Okun's law, and translate an unemployment forecast into an inflation forecast.

SECTION 01

Demand, supply, elasticity

Demand slopes down and supply slopes up because buyers face diminishing marginal utility while sellers face rising marginal cost

Market equilibrium is the (P^, Q^*) pair at which the quantity buyers demand equals the quantity sellers supply — the starting point for every market structure that follows.*

At equilibrium: $Q_D(P^*) = Q_S(P^*) \rightarrow$ no surplus, no shortage

where: Shifts in either curve move P^ and Q^* in predictable directions; elasticity determines which of the two moves more.*

LAW OF DEMAND

Holding income, prices of related goods, and preferences constant, higher $P \Rightarrow$ lower Q_D . Two drivers: substitution effect (relative price up) and income effect (real purchasing power down).

LAW OF SUPPLY

Holding input prices and technology constant, higher $P \Rightarrow$ higher Q_S . Driver: rising marginal cost beyond some threshold — producers need a higher price to cover MC on the next unit.

SHIFTS vs MOVEMENTS

A price change MOVES along the curve; a change in income, input prices, technology, or expectations SHIFTS the whole curve. Students confuse this on day one — do not.

Source: Prerequisite microeconomics. CFA Institute, Curriculum Vol 2, Learning Module 1 assumes this framework (Introduction, pp. 3–4).

Own-price elasticity measures how responsive quantity demanded is to price — and it determines whether cutting price raises or lowers revenue

Revenue rule: cut price if demand is elastic, raise price if demand is inelastic, and if it is unit-elastic you are already sitting on the revenue maximum.

$$E_d = (\% \Delta Q_D) / (\% \Delta P)$$

$$MR > 0 \Leftrightarrow \text{elastic}$$

$$MR < 0 \Leftrightarrow \text{inelastic}$$

where: $|E_d| > 1$ elastic · $|E_d| = 1$ unit-elastic (TR peak, $MR = 0$) · $|E_d| < 1$ inelastic. That MR chain is the CFA's own link between the demand curve and the total-revenue curve.

WORKED EXAMPLE — ELASTIC DEMAND

A firm cuts price from 10.00 to 9.00 (−10%) and observes quantity sold rise from 100 to 120 (+20%). What is E_d , and what happens to revenue?

1. $E_d = (+20\%) / (-10\%) = -2.0 \Rightarrow |E_d| = 2 > 1$, demand is ELASTIC.
2. Old TR = $10 \times 100 = 1,000$; New TR = $9 \times 120 = 1,080$.
3. TR rose by 80 (+8%) because the quantity effect dominated the price effect.
4. Same cut with $|E_d| = 0.5$ (inelastic): Q rises only 5% to 105, TR = $9 \times 105 = 945$ — revenue FALLS.

$E_d = -2.0 \rightarrow \text{elastic} \rightarrow \text{cut price, revenue rises.}$

¹ CFA materials usually quote E_d as an absolute value. Keep the minus sign for algebraic work — the law of demand makes it negative.

² Elasticity $\neq$ slope: slope is $\Delta Q / \Delta P$, elasticity is $(\Delta Q / Q) / (\Delta P / P)$. A linear demand curve has constant slope but variable elasticity.

Source: CFA Institute, Curriculum Vol 2, LM 1 Overview p. 3 (three elasticities) and pp. 6–7 (TR rises where $MR > 0$). Worked numbers illustrative.

Cross-price and income elasticities separate substitutes from complements and normal goods from inferior — the three-elasticity triad for any demand forecast

A demand forecast that does not specify the three elasticities is not a forecast — it is an assertion with no stress-test.

Elasticity	Formula	Sign	Interpretation & example
Own-price (E_d)	$(\% \Delta Q_x) / (\% \Delta P_x)$	Always < 0	Slopes down; gasoline ≈ -0.3 (inelastic).
Cross-price, substitutes	$(\% \Delta Q_x) / (\% \Delta P_y)$	Positive	P_y up $\Rightarrow Q_x$ up — Pepsi vs Coca-Cola.
Cross-price, complements	$(\% \Delta Q_x) / (\% \Delta P_y)$	Negative	P_y up $\Rightarrow Q_x$ down — printers vs ink.
Income, normal good	$(\% \Delta Q_x) / (\% \Delta I)$	Positive	Income up $\Rightarrow Q_x$ up — restaurant meals.
Income, luxury	$(\% \Delta Q_x) / (\% \Delta I)$	> 1	Wallet share rises — fine wine, couture.
Income, inferior good	$(\% \Delta Q_x) / (\% \Delta I)$	Negative	Income up $\Rightarrow Q_x$ down — instant noodles.

WHAT SIGNALS A STRUCTURAL SHIFT

A rise in income elasticity for a staple (coffee becoming a "luxury" single-serve SKU) or a sign-change in cross-elasticity (streaming turning into a SUBSTITUTE for cinema rather than a complement) both signal that a category boundary is being redrawn — and that incumbents must re-price before the market re-prices them for it.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1 Overview, p. 3 (own-price, cross-price, income elasticity). Signs, formulas and examples are standard Level I background.

SECTION 02

Firm costs & profit maximization

Economic profit subtracts the opportunity cost of every factor — a firm with positive accounting profit can still be destroying economic value

Normal profit = zero economic profit = a rate of return on capital exactly equal to what an equivalently risky alternative investment would have earned. It is payment for showing up, not a reward for excellence.

$$\text{Economic profit} = \text{TR} - (\text{Accounting cost} + \text{Opportunity cost})$$

where: Accounting profit = TR – accounting costs. Economic cost considers the total opportunity cost of ALL factors of production, capital included.

THE ENTREPRENEUR EXAMPLE

Founder nets USD 120,000 in accounting profit from her consultancy. She turned down a corporate role paying USD 150,000 to start the firm.

Accounting profit = +120,000.

Economic profit = 120,000 – 150,000 = –30,000.

She is destroying value. Zero economic profit is the bar, not zero accounting profit.

WHY IT MATTERS FOR INVESTORS

In a competitive long-run equilibrium with free entry, firms earn ZERO economic profit — returns just compensate for risk. Positive economic profit is therefore evidence of a barrier: a moat protecting rents. The analyst's job is to price how long that barrier holds.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Overview p. 4 and "Breakeven Analysis and Shutdown Decision", p. 9.

A firm maximizes profit where marginal revenue equals marginal cost and marginal cost is not falling — the single rule that governs every market structure

Under perfect competition $MR = P$ (horizontal demand curve). Under every imperfectly competitive structure $MR < P$, because selling one more unit requires cutting the price on all units. The rule itself never changes.

Maximize profit $\Leftrightarrow MR = MC$ AND MC not falling with output

where: First-order condition: slope of TR equals slope of TC. Second-order condition: MC must cut MR from below — where $P = MC$ on a falling MC curve you are at a profit MINIMUM.

CFA LM 1 EXAMPLE 1 — WR INTERNATIONAL, IMPERFECT COMPETITION

Prefab-housing firm facing a downward-sloping demand curve. Quantity moves in blocks of 10; price falls 250 per block. Where does it break even, where is the profit region, and where is profit maximised?

1. Initial breakeven ($TR = TC$): $Q = 40$ at $P = 9,000 \rightarrow TR = TC = 360,000$.
2. Region of profitability: $40 < Q < 80$. At $Q = 80$, $TR = TC = 640,000$ (upper breakeven).
3. Profit maximum: $Q = 60$, $P = 8,500 \rightarrow TR = 510,000$, $TC = 480,000 \rightarrow \text{profit} = 30,000$.
4. Check MR vs MC: from $Q 50 \rightarrow 60$, $MR = 7,250 > MC = 6,000$; from $60 \rightarrow 70$, $MR = 6,750 < MC = 7,000$. The crossing sits at $Q = 60$.

$Q^* = 60$, $P^* = 8,500$, maximum economic profit = 30,000.

¹ CFA trap: the profit maximum is NOT where average total cost is minimised. Those two quantities coincide only by accident.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, Example 1 (WR International), pp. 11–12; profit-maximization conditions p. 4 and pp. 7–8.

Breakeven, shutdown, and long-run exit are three distinct thresholds defined by whether price covers total cost, variable cost only, or nothing

Short run: at least one factor is fixed, fixed costs are sunk, and only variable costs drive the decision. Long run: every factor is variable, so the firm must cover ATC to justify staying.

Revenue vs cost (CFA Exhibit 7)	Short-run decision	Long-run decision
$TR > TC$ ($P > ATC$)	Operate – econ. profit > 0	Stay in market
$TR = TC$ ($P = \min ATC$) breakeven	Operate – normal profit	Stay in market
$TVC < TR < TC$ ($\min AVC < P < ATC$)	Operate – loss $< TFC$	Exit market
$TR = TVC$ ($P = \min AVC$) shutdown pt	Indifferent – loss = TFC	Exit market
$TR < TVC$ ($P < \min AVC$)	Shut down – loss capped at TFC	Exit market

CFA EXAMPLE 2 – THE LONDON BUSINESS

Revenue GBP 2m, TC GBP 2.5m (TFC 1m + TVC 1.5m) → net loss 500k. Shutting down would cost the full 1m of fixed cost, so operating minimises the loss: keep going and renegotiate with creditors. Change revenue to GBP 1.3m and the loss becomes 1.2m — now revenue no longer covers TVC, so shutting down caps the loss at 1m and saves shareholders 200k.

¹ Sunk fixed costs are IGNORED in the short-run decision. Economists call min AVC the shutdown point and min ATC the breakeven point.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Breakeven Analysis and Shutdown Decision" and Exhibit 7, pp. 9–13.

Economies of scale pull long-run average cost down until minimum efficient scale; diseconomies push it back up — the classic U-shape of LRAC

LRAC is the envelope curve of every possible short-run ATC curve. Its minimum point is minimum efficient scale — per the CFA text, the optimal firm size under perfect competition in the long run.

DRIVERS OF ECONOMIES OF SCALE

- Increasing returns to scale — output rises proportionately faster than inputs.
- Division of labor and specialization — each worker becomes proficient at a narrow task.
- Affording more expensive but more efficient equipment and the latest technology.
- Bargaining power — discounted input prices when buying in larger quantities.
- Less waste, marketable by-products, better market information for decision-making.

CFA examples: electric utilities (capacity), Walmart (bulk purchasing + point-of-sale data).

DRIVERS OF DISECONOMIES OF SCALE

- Decreasing returns to scale — output rises proportionately less than inputs.
- Being so large the firm cannot be properly managed — bureaucracy and communication breakdown.
- Overlapping and duplicating business functions and product lines.
- Higher input prices from supply constraints when buying in very large quantities.

CFA example: pre-restructuring GM — overlapping models, geographic coordination failure, above-market labour cost. Exited three brands in 2009 and sold Opel in 2017.

Minimum efficient scale (MES) = the output at which LRAC reaches its minimum. Below MES a firm is sub-scale and must grow; beyond MES it is in diseconomies and must shrink production.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Economies and Diseconomies of Scale", Exhibits 8–12 and Example 3, pp. 14–19.

SECTION 03

Four market structures

Five determinants separate the four market structures — sellers, differentiation, barriers, pricing power, and non-price competition

Quantity sold is highest and price usually lowest under perfect competition. But the CFA is careful: less-competitive structures can fund R&D that perfect competition would bankrupt — consumers do not always lose.

Structure	Sellers	Product	Barriers to entry	Pricing power	Non-price competition
Perfect competition	Many	Homogeneous	Very low	None	None
Monopolistic competition	Many	Differentiated	Low	Some	Advertising + differentiation
Oligopoly	Few	Homogeneous	High	Some or considerable	Advertising + differentiation
Monopoly	One	Unique	Very high	Considerable	Advertising

PORTER'S FIVE FORCES — THE PRACTITIONER'S OVERLAY (CFA, p. 24)

Academic theory asks "how many sellers?"; Porter asks threat of entry, power of suppliers, power of buyers, threat of substitutes, and rivalry among incumbents. Four of the five map onto the grid above. The orphan — power of suppliers — is the one economic theory under-weights and practitioners use to identify a durable "economic moat": no credible entrant, and customers locked in by switching costs.

¹ Exhibit 13 files oligopoly product as homogeneous/standardized, though in practice it may be branded (cereals) or not (cement).

² Long-run economic profit: zero under perfect and monopolistic competition (free entry); positive only where barriers hold entrants out.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Introduction to Market Structures", Exhibit 13, pp. 20–24.

In perfect competition price equals marginal cost and economic profit is zero in the long run — the benchmark every other structure is measured against

No firm has pricing power, so the demand curve each individual firm faces is perfectly elastic (horizontal) at the market price — $P = MR = AR$.

Short run: $P = AR = MR = MC$

Long run: $P = MC = \min(ATC)$

where: Free entry drives long-run price down to minimum ATC, at which point economic profit is zero and each firm earns only normal profit.

INDUSTRY FEATURES

- Many small sellers, each a price taker.
- Identical, undifferentiated output.
- Perfect information.
- No entry or exit costs.
- No advertising — nothing to differentiate.

SHORT-RUN ECON. PROFIT

Possible whenever the firm's ATC sits below the market price. But the excess attracts entrants, output rises, and market price is driven down to every firm's minimum ATC. Short run: any outcome. Long run: normal profit only.

REAL-WORLD PROXIES

The CFA insists this is not just an ideal: commodity agriculture (the Nebraska hog farmer, the Bavarian milk producer), spot FX in major pairs. Millions of businesses do very well under exactly this pressure.

¹ Only perfect competition has a well-defined supply function — the MC schedule. Elsewhere, output comes from $MR = MC$, price from demand.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, perfect-competition analysis, Exhibits 1–5, pp. 6–13 and pp. 20–21.

Monopolistic competition combines many sellers with product differentiation — firms face downward-sloping demand but still earn zero long-run economic profit

Differentiation buys short-run pricing power; low entry and exit costs eventually compete it away. Chamberlin and Robinson named it precisely because it is a hybrid.

Short run: set Q where $MR = MC$, read P off the demand curve $\Rightarrow$
 $\pi = (P - ATC) \times Q$

where: Long run: entry pulls each firm's demand curve down until it is tangent to ATC, so $P = ATC$ and $\pi = 0$ — but at Q_1 below the Q_2 that minimises average cost. That gap is the excess capacity of monopolistic competition.

WHY IT DOMINATES REAL MARKETS

Most consumer markets sit here: soft drinks, cosmetics, apparel, phones, cafés. The CFA's defence of it is Schumpeterian — differentiation is the profit incentive to innovate, and consumers demonstrably value variety (Germany and Japan sell each other cars).

THE BRAND MOAT

The more successful the differentiation, the more the firm resembles a monopoly and the longer margins survive — Coca-Cola, Harley-Davidson. The cost is real too: advertising is an economic cost that perfect competitors never incur.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Monopolistic Competition", Exhibits 14–15, pp. 25–28.

Non-colluding oligopolists have three pricing strategies — and the kinked demand curve explains why oligopoly prices get stuck

The asymmetry that drives everything: rivals MATCH a price cut but IGNORE a price rise. That bends the demand curve at the prevailing price and tears a gap in marginal revenue.

1 • INTERDEPENDENCE

Assume rivals match any price cut and ignore any price rise. Demand is then highly elastic ABOVE the prevailing price (customers leave for cheaper rivals) and inelastic BELOW it (rivals follow, so you gain little volume). Result: a kinked demand curve.

2 • COURNOT

Each firm picks its own output assuming the rival's quantity will not change. Solve the two reaction equations simultaneously.

Worked through in full on the next slide.

3 • NASH

Each firm best-responds to what it expects the rival to do; in equilibrium nobody can gain by deviating alone.

Two slides on — and the answer is not the cell most people guess.

WHY THE KINK MAKES PRICES STICKY — AND WHY THE MODEL IS INCOMPLETE

The kink produces a DISCONTINUOUS marginal revenue curve: a vertical gap at the prevailing quantity. Any marginal cost curve passing through that gap gives the same profit-maximising price and output, so costs can move a long way before the firm re-prices. That is why oligopoly prices are observed to be stable. The criticism: the model explains why the prevailing price persists but cannot say what set it, so the CFA calls it an incomplete pricing analysis.

¹ Beyond the three: a dominant firm (CFA threshold ~40% share) can act as price leader, and open, formal collusion turns the group into a cartel.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Demand Analysis and Pricing Strategies in Oligopoly Markets", Exhibits 16–17, pp. 29–32.

The Cournot duopoly assumes each firm treats its rival's output as fixed — solving the two reaction curves gives a price between competition and monopoly

Cournot (1838) predates game theory but delivers the same answer as the Nash equilibrium of the quantity-setting game — each firm best-responds to the rival's output.

$$\text{Setup: } Q_D = 450 - P, \quad MC = 30, \quad Q_D = q_1 + q_2$$

where: Each firm maximises its own profit assuming the rival's quantity is fixed. Two reaction equations, two unknowns, solve simultaneously.

CFA LM 1 — DUOPOLY WITH LINEAR DEMAND

Two firms choose quantities q_1 and q_2 with $P = 450 - (q_1 + q_2)$. Each sets $MR = MC = 30$ taking the other's quantity as given.

1. $TR_1 = (450 - q_1 - q_2) \cdot q_1 \rightarrow MR_1 = 450 - 2q_1 - q_2 = 30$.
2. By symmetry: $450 - q_1 - 2q_2 = 30$. With $q_1 = q_2$: $450 - 3q_1 = 30 \rightarrow q_1 = q_2 = 140, Q = 280$.
3. Price: $P_{\text{Cournot}} = 450 - 280 = 170$.
4. Benchmarks: perfect competition sets $P = MC \rightarrow 450 - Q = 30 \rightarrow Q = 420, P = 30$. Monopoly sets $MR = MC$ on market demand $\rightarrow 450 - 2Q = 30 \rightarrow Q = 210, P = 240$.

Cournot: $Q = 280, P = 170$ — squarely between competition (30) and monopoly (240).

¹ As the number of firms rises, the Cournot outcome converges toward competition — historically the basis for US antitrust policy.

² Contrast Stackelberg: Cournot moves are simultaneous, Stackelberg sequential — the first mover earns more, the follower less.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "The Cournot Assumption" and Exhibit 18, pp. 32–33.

In a Nash equilibrium no firm can improve its own profit by unilaterally changing strategy — even when a cooperative outcome would pay both of them more

Payoffs are (ArcCo, BatCo). Find each firm's best response to each rival strategy — the cell where BOTH are best-responding is the Nash equilibrium.

ArcCo strategy \ BatCo strategy	BatCo — LOW price		BatCo — HIGH price	
	ArcCo	BatCo	ArcCo	BatCo
ArcCo — LOW price	50	70	80	0
ArcCo — HIGH price	300	350	500	300

READING THE PAYOFF MATRIX — SOLVE IT COLUMN BY COLUMN

Each firm picks Low or High simultaneously. Find each firm's dominant strategy, then the Nash equilibrium and the cooperative equilibrium.

1. BatCo: if ArcCo LOW, $70 > 0 \Rightarrow$ play LOW. If ArcCo HIGH, $350 > 300 \Rightarrow$ play LOW. LOW is dominant for BatCo.
2. ArcCo: if BatCo LOW, $300 > 50 \Rightarrow$ play HIGH. If BatCo HIGH, $500 > 80 \Rightarrow$ play HIGH. HIGH is dominant for ArcCo.
3. Nash equilibrium = (ArcCo HIGH, BatCo LOW), payoffs $300 + 350 = 650$. Neither can gain by moving alone.
4. Joint profit is maximised at (HIGH, HIGH) = $500 + 300 = 800$. BatCo will only go there if ArcCo shares at least 51 of its 500 — $300 + 51 > 350$.

Nash = (ArcCo HIGH, BatCo LOW) = 650. Cooperative = (HIGH, HIGH) = 800. Side-payment needed: ≥ 51 .

¹ Collusion succeeds when sellers are few, products homogeneous, cost structures similar, orders small and frequent, retaliation severe.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "The Nash Equilibrium", Exhibit 19 (ArcCo / BatCo), pp. 34–36.

A single-price monopolist sets MR equal to MC on the whole market demand curve — producing less and pricing higher than competition

The monopolist finds quantity from $MR = MC$ and then charges whatever the demand curve supports at that quantity. Lower output plus higher price is the CFA's core comparison with perfect competition.

$$\text{Same setup: } P = 450 - Q, \quad MC = 30 \quad \rightarrow \quad MR = 450 - 2Q$$

where: $TR = P \cdot Q = 450Q - Q^2$, so $MR = 450 - 2Q$ — MR falls twice as fast as demand. Set $MR = MC$: $450 - 2Q = 30 \rightarrow Q = 210$; then read price off demand: $P = 450 - 210 = 240$.

Market structure	Quantity	Price	Consumers	Producers
Perfect competition ($P = MC$)	420	30	Best	Normal profit
Cournot duopoly (two firms)	280	170	Intermediate	Positive
Monopoly ($MR = MC$)	210	240	Worst	Maximum

WHERE MONOPOLY POWER ACTUALLY SHOWS UP

You rarely meet a pure monopolist; you do meet dominant firms — the CFA puts the threshold near 40% share — acting as price leaders while followers accept the price rather than fight a lower-cost producer. The other upgrade is price discrimination: different prices to different buyers (fare buckets, student pricing, volume tiers) turns consumer surplus into profit.

¹ Price discrimination is an extension beyond LM 1, included here because it is where monopoly rent actually shows up in practice.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, monopoly benchmark within the Cournot discussion and Exhibit 18, pp. 32–33; dominant-firm price leadership pp. 36–37.

Concentration ratios and the Herfindahl-Hirschman Index measure how concentrated a market is — and the HHI catches the mergers a concentration ratio ignores

HHI squares market shares before summing, so big firms count disproportionately. M equal firms give $HHI = 1/M$ — five equal firms score 0.20, a monopolist scores 1.00.

$$CR_N = \sum_{i=1}^N s_i$$

$$HHI = \sum_i s_i^2$$

where: s_i = market share of firm i as a decimal. CR_N ranges from 0 (perfect competition) to 100% (monopoly); HHI from ~0 to 1.00. Both are normally quoted on the top N firms.

CFA LM 1 — FIVE-FIRM MARKET (ZETA / YUSEF / XENON / WATERS / VLASTOS)

Shares: Zeta 35%, Yusef 25%, Xenon 20%, Waters 10%, Vlastos 10%. Compute CR_3 and HHI_3 , then re-run both after the top two merge.

1. $CR_3 = 35 + 25 + 20 = 80\%$ — the top three control four-fifths of the market.
2. $HHI_3 = 0.35^2 + 0.25^2 + 0.20^2 = 0.1225 + 0.0625 + 0.0400 = 0.2250$.
3. Merger: Zeta + Yusef $\rightarrow$ 60%. CR_3 moves only from 80% to $60 + 20 + 10 = 90\%$ (+10 pp).
4. HHI_3 goes to $0.60^2 + 0.20^2 + 0.10^2 = 0.36 + 0.04 + 0.01 = 0.4100$ — an 82% jump on a barely-moved CR.

Pre-merger $HHI_3 = 0.225 \rightarrow$ post-merger 0.410. HHI is the merger-sensitive gauge; CR is not.

¹ Neither measure captures barriers to entry or demand elasticity — a lone incumbent facing a credible entrant still prices competitively.

² US screens: the 2010 guidelines used HHI 1,500 / 2,500; the 2023 guidelines presume harm above HHI 1,800 with an increase over 100.

Source: CFA Institute, Curriculum Vol 2, Learning Module 1, "Determining Market Structure", concentration-ratio box, Example 4 and question set, pp. 40–43.

SECTION 04

Business cycle phases & indicators

The business cycle has four phases — recovery, expansion, slowdown, contraction — distinguished by the sign and direction of the output gap

Output gap = actual GDP – potential GDP. Two questions define the phase: is the gap positive or negative, and is it widening or narrowing? Recovery and expansion close then overshoot; slowdown and contraction unwind.

Phase	Output gap	Employment	Inflation
Recovery	Negative, starting to narrow	Layoffs slow; overtime before hiring	Remains moderate
Expansion	Turns positive – boom	Overtime → hiring; U stabilises, falls	Picks up modestly
Slowdown	Peak positive gap, narrowing	Hiring continues, slower; U still falls	Accelerates further
Contraction	Turns negative	Hours cut, hiring freeze, then layoffs	Decelerates with a lag

RULE OF THUMB vs THE NBER METHOD

The informal rule: two consecutive quarters of negative real GDP growth = recession. The CFA calls this misleading — a slightly positive middle quarter between two negatives breaks it. The NBER Business Cycle Dating Committee instead reads a basket (employment, industrial production, sales) and applies Burns & Mitchell principles, which is why its determinations often land well after the turning point itself. Useful rule for a real-time call; wrong tool for a definitive date.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Phases of the Business Cycle" and Exhibit 5, pp. 52–55.

Classical, growth, and growth-rate cycles are three different measuring sticks — most modern analysts use the growth cycle around trend GDP

The CFA is explicit: the classical cycle is rarely used, and it will generally use the growth-cycle concept. Know all three anyway — practitioners use the words interchangeably and cause exactly the confusion you are about to avoid.

CLASSICAL CYCLE

Fluctuations in the LEVEL of economic activity (volume GDP).

Contractions between peaks and troughs are short; expansions are much longer.

Rarely used, for two CFA-stated reasons: it does not separate short-term fluctuation from long-run trend, and outright absolute declines in activity are infrequent.

GROWTH CYCLE

Fluctuations AROUND the trend or potential growth path — that is, the output gap.

Peaks come earlier and troughs later than in the classical cycle; time above and below trend is of similar length.

Closest to how mainstream economists think. The concept used in this deck.

GROWTH-RATE CYCLE

Fluctuations in the GROWTH RATE itself (e.g. YoY or annualised MoM GDP growth).

Advantage: no need to estimate a long-run trend first.

Peaks and troughs are recognised EARLIEST of the three — which is why central-bank watchers phrase calls as "growth has peaked".

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Types of Cycles" and Exhibits 2–4, pp. 52–54.

Credit cycles are longer and deeper than business cycles and amplify them — loose credit funds the excesses that later unwind

The symmetry matters both ways: recessions accompanied by house- and equity-price busts are longer and deeper, and recoveries accompanied by rapid credit, risk-taking and house-price growth are stronger.

WHAT CREDIT CYCLES MEASURE

The changing AVAILABILITY and PRICING of credit — growth in private-sector credit. BIS composite proxy: real credit, credit-to-GDP ratio, real house prices. Market proxies: bank lending standards, high-yield spreads, LTVs on new mortgages.

WHY THEY AMPLIFY

Credit finances construction and property purchase. Rising asset prices raise household wealth and collateral values, which supports more credit. When lenders pull back, the unwind is self-reinforcing: falling values → defaults → bank losses → further tightening. Strong credit-cycle peaks are closely associated with subsequent systemic banking crises.

Crisis (CFA-cited)	Region	Credit-cycle precursor
Latin America debt crisis, 1980s	Latam	Sovereign USD borrowing boom, then commodity bust
Mexico, Brazil, Russia crises, 1990s	Emerging	External debt build-up; capital-flow reversal
Asian financial crisis, 1997–1998	East Asia	Property and corporate USD-lending boom
Global Financial Crisis, 2008–2009	Global	US subprime mortgage leverage and securitisation

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Credit Cycles" and BIS Exhibit 6, pp. 57–60.

Leading, coincident, and lagging indicators each answer a different question — what will happen, what is happening, what just happened

A phase call uses all three: a coincident indicator to anchor the present, a leading one to extrapolate, a lagging one to confirm the last turn. No single indicator is definitive — that is the CFA's own answer.

Category	Example indicators (CFA Exhibit 10)	Use
Leading	Stock market, house prices, retail sales, interest spread (LT vs ST), building permits, consumer expectations, average weekly hours (mfg), manufacturers' new orders	Predict the next turn
Coincident	Industrial production index, real personal income, manufacturing and trade sales, non-farm payrolls	Identify the current phase
Lagging	Average duration of unemployment, inventory-to-sales ratio, change in unit labour cost, inflation, average prime lending rate, consumer instalment debt / income, C&I loans outstanding	Confirm the previous turn

DIFFUSION INDEX — READ BREADTH, NOT JUST LEVEL

A diffusion index is the proportion of a composite's components moving consistently with the overall index. The Conference Board scores each of the 10 LEI components 1.0 if it rises more than 0.05%, 0.5 if it moves less than that, 0 if it falls more than 0.05% — then averages and multiplies by 100. A composite rising because 8 of 10 components are up is a far stronger signal than one dragged up by two outliers. Separately, nowcasting uses high-frequency data (card spend, search, transactions) to estimate the present before official data lands.

¹ Conference Board LEI, components 1–5: ISM new orders · average weekly hours (mfg) · initial jobless claims · new orders, consumer goods.

² LEI 6–10: new orders, non-defence capital goods ex-aircraft · building permits · S&P 500 · Leading Credit Index · 10y–fed-funds spread · consumer expectations.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Economic Indicators", Exhibits 10–12 and the diffusion-index box, pp. 65–72.

SECTION 05

Unemployment & inflation

The unemployment rate counts the jobless who are actively looking for work — on a labor-force base that itself moves with the cycle

The CFA reason unemployment lags: hiring and firing are both costly, so businesses delay both. Unemployment rises slowly as recessions start and falls slowly as recoveries begin.

$$U = \text{Unemployed} / \text{Labor force} \qquad \text{LFPR} = \text{Labor force} / \text{Working-age population}$$

where: Labor force = employed + unemployed and actively seeking. It excludes discouraged workers who have stopped searching, and the institutionalised population.

Component	Definition	Cycle behavior
Frictional	Short-term job transitions and search friction	Roughly stable
Structural	Skill or geographic mismatch with vacancies	Slow-moving
Cyclical	Aggregate-demand driven – rises in recession	Highly cyclical
Natural rate (NAIRU)	Frictional + structural; stable-inflation rate	Slow-moving

Both edges cut the same way: the headline U-rate can FALL early in a downturn as discouraged workers leave the labor force, and RISE early in a recovery as job seekers re-enter (CFA LM 2, PP 15). Always read U with the LFPR.

Source: Illustrative — standard Level I macroeconomics framework. Lagging-indicator behaviour is CFA Vol 2, LM 2, Exhibit 10 and practice problems 14–15, pp. 66 and 76.

CPI, core CPI, PPI, and the GDP deflator measure inflation differently – and they can disagree by meaningful margins

Central banks target core measures because food and energy are volatile supply-driven components that monetary policy cannot address without wrecking employment on the way.

Measure	Basket / scope	Weights	Typical use
Headline CPI	Urban household consumption basket	Laspeyres (fixed)	Cost-of-living index
Core CPI	CPI excluding food and energy	CPI ex-volatiles	Underlying trend
PPI	Producer selling prices at wholesale	Output-weighted	Early-stage signal
GDP deflator	All domestically produced goods/services	Paasche (current)	Whole-economy prices
PCE deflator (US)	Personal consumption expenditure basket	Chained, current	Fed core gauge

WHY HEADLINE AND CORE DIVERGE – AND WHY IT IS NOT A CONTRADICTION

An oil-price spike lifts headline CPI immediately but raises core CPI only to the extent it leaks into transport, heating and the input costs of non-energy goods. Central banks "look through" the first effect and react only to the second. That is why "CPI print high, central bank unmoved" is a coherent story rather than a policy error. The GDP deflator can also diverge for a structural reason: it covers domestic PRODUCTION, so imported price moves hit CPI but not the deflator.

Source: Illustrative — standard Level I macroeconomics framework. Central-bank practice per CFA Vol 2, LM 4 (the Fed watches core PCE around 2%), p. 121.

The Phillips curve maps the short-run trade-off between inflation and unemployment — but the long-run curve is vertical at the natural rate

Short run: the central bank can buy less unemployment by accepting more inflation. Long run: it cannot — it gets the inflation and keeps the same unemployment.

$$\begin{array}{ll} \text{Short run:} & \pi = E[\pi] - \beta (U - U_{\text{natural}}) \\ & = U_{\text{natural}} \quad (\text{vertical}) \end{array} \qquad \text{Long run:} \quad U$$

where: π = inflation; $E[\pi]$ = expected inflation; U = unemployment rate; U_{natural} = NAIRU; $\beta > 0$ is the short-run slope. Note the sign: U above NAIRU pushes π below expectations.

SHORT RUN

With expectations fixed, U and π trade off along one curve. A demand surprise pushing U below NAIRU raises π above $E[\pi]$. Central banks exploit this briefly to "run the economy hot" — the trade-off is real, just temporary.

LONG RUN

Agents update. If actual π exceeds expected π period after period, $E[\pi]$ rises and the whole short-run curve shifts UP — same U , higher π . Stack those shifts and the long-run locus is vertical at NAIRU.

That is the Friedman–Phelps result, and it is why anchored expectations are a central bank's most valuable asset.

Source: Illustrative — standard Level I macroeconomics; Phillips (1958) with the Friedman–Phelps expectations-augmented extension.

Okun's law ties the output gap to the unemployment gap at roughly 2-to-1 — a rule of thumb that holds across most developed economies

Okun's law is the bridge from a GDP forecast to an unemployment forecast — and, once combined with Phillips, all the way to an inflation forecast.

$$(Y - Y^*) / Y^* \approx -\alpha \times (U - U_{\text{natural}}), \quad \alpha \approx 2.0$$

where: A 1 pp rise in the unemployment rate above NAIRU corresponds to output roughly 2% below potential. The minus sign is the whole content of the law.

ILLUSTRATIVE — MAPPING U-GAP TO OUTPUT GAP

Economy with potential growth 2% p.a. and NAIRU = 4%. Observed unemployment rises from 4% to 6% over the year. Estimate the output gap.

1. Unemployment gap: $U - U_{\text{natural}} = 6\% - 4\% = +2$ pp.
2. Okun with $\alpha = 2.0$: output gap $\approx -2.0 \times 2$ pp = -4%.
3. Interpretation: actual GDP is running about 4% below potential — recessionary territory.
4. Run it backwards: closing 3 pp of that output gap would absorb roughly 1.5 pp of the unemployment gap.

U-gap +2 pp → output gap $\approx -4\%$ (Okun coefficient 2.0).

¹ The coefficient varies by country and era: post-1990 US near 2.0; Japan far higher, as employment norms make output absorb the shock.

Source: Illustrative — Okun (1962), original US coefficient ≈ 2.0 ; roughly 1.5–2.5 across developed markets.

Combining Phillips and Okun lets an analyst translate an unemployment forecast into an inflation forecast — a core input for any rate-call narrative

This is the quick model central-bank watchers run in their heads: unemployment gap $\Rightarrow$ output gap $\Rightarrow$ inflation gap $\Rightarrow$ policy stance.

ILLUSTRATIVE — RECESSION SCENARIO

Starting state: $U = U_{\text{natural}} = 4\%$, $\pi = E[\pi] = 2\%$. Shock: a demand-driven recession pushes U to 6%. Assume Okun $\alpha = 2.0$ and Phillips $\beta = 0.5$.

1. Step 1 — Unemployment gap: $U - U_{\text{natural}} = 6\% - 4\% = +2$ pp.
2. Step 2 — Okun output gap: -2.0×2 pp = -4% ; actual GDP sits 4% below potential.
3. Step 3 — Phillips: $\pi = E[\pi] - \beta \times (U - U_{\text{natural}}) = 2\% - 0.5 \times 2 = 1\%$.
4. Step 4 — Policy reading: a 100 bp inflation undershoot alongside a -4% output gap is a clear case for easing.

π falls from 2% to 1% via the short-run Phillips trade-off; output gap = -4% .

WHERE THIS MODEL BREAKS — SAY IT BEFORE SOMEONE ELSE DOES

The short-run Phillips slope β has flattened materially since the 1990s as inflation expectations became well anchored, so inflation responds far less to the unemployment gap than older textbooks imply. And supply shocks — energy, pandemic-era supply chains — push π and U in the SAME direction, which breaks the trade-off outright. That is precisely why the 2021–2023 episode confounded the simple model.

Source: Illustrative — applies the Phillips and Okun relationships from the two preceding slides to a stylised scenario. No CFA worked example is cited.

SECTION 06

Investment-relevant cycle dynamics

Equity markets trough three to six months before the economy does, and credit and the curve turn earlier still — which is why both sit inside the LEI

Equities discount future cash flows, so they turn before reported earnings. Credit discounts future default probability, so it turns before equity fundamentals confirm anything.

EQUITY MARKETS

The CFA is specific: equity markets typically hit a TROUGH about three to six months before the economy bottoms out, and well before indicators turn up.

The S&P 500 is one of the 10 Conference Board LEI components for exactly this reason.

Take-away: by the time GDP confirms a turn, equities have already priced it.

YIELD CURVE

The LEI component is the spread between 10-year Treasury yields and the fed funds rate.

Inversion means short rates are expected to FALL — the market pricing a cutting cycle to defend against weakness.

In US data it has preceded every recession since the late 1960s, typically by 12–24 months, with a small number of false positives. A long and variable lead, not a timing tool.

CREDIT SPREADS

The Leading Credit Index — six financial indicators reflecting the financial system's ability to absorb stress — is also an LEI component.

Spreads tend to be tightest mid-cycle and widen through the late cycle as underwriting standards deteriorate.

The CFA logic: a vulnerable financial system amplifies negative shocks into broad recessions.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, "Market Conditions and Investor Behavior", p. 55, and Exhibits 10–11, pp. 66–67.

Inventory and capital-spending cycles amplify the real economy — light equipment orders move first, heavy capex last

Watch the inventory-to-sales RATIO, not the inventory level: it falls in recovery, is stable through expansion, rises in slowdown as sales decelerate ahead of production, then falls back to normal as contraction clears the excess.

Phase	Inventory / sales ratio	Capital spending	Orders placed
Recovery	FALLS – sales recover faster than production	Low but rising; efficiency focus, not capacity	Software, IT hardware
Expansion	STABLE – restocking keeps pace with sales	Capacity expansion; strong cash flow funds capex	Heavy equipment, factories
Slowdown	RISES – sales slow faster than production	Orders continue at or near full capacity – a late-cycle tell	Capacity, e.g. TMT fibre
Contraction	FALLS back to normal – output cut below sales	Orders halted, some in-flight orders cancelled; maintenance cut	Light/tech cut first

WHY THIS DRIVES SECTOR SELECTION

Order lead times set the sequence. Recovery capex goes to high-obsolescence kit — software, systems, hardware — so technology and semis lead early. Expansion capex funds heavy and complex equipment, warehouses and factories. In a downturn the sequence reverses: light equipment and technology are cut first because those orders are easy to cancel, while long-planned construction runs on and is cut last. Any sector view should ask where the industry sits on that order-book lead-lag curve.

¹ Inventories are small relative to GDP but swing hard. Capital spending is one of the most procyclical, volatile components of GDP.

Source: CFA Institute, Curriculum Vol 2, Learning Module 2, Exhibit 8 (capital spending) and Exhibit 9 (inventories), pp. 61–64.

Sector rotation maps cycle phases to portfolio tilts — cyclicals and risk assets early, defensives and duration late

The CFA gives the skeleton: risky assets are repriced upward when markets anticipate expansion, and safe assets — government bonds, utilities, staple producers — are prized in contraction. The sector detail below is practitioner convention layered on top.

Phase	Equity style / size	Favored sectors	Fixed income
Recovery	Small caps, value, cyclicals	Consumer discretionary, financials, homebuilders	HY credit rally
Expansion	Mid-cap quality, growth	Industrials, technology, materials	IG credit tight
Slowdown	Large-cap quality, low volatility	Energy, late-cycle industrials, commodities	Duration extends
Contraction	Defensives, long duration bonds	Consumer staples, utilities, healthcare	Government bonds

LIMITATIONS — DO NOT MISTAKE THIS FOR A FORECAST

Sector rotation works on average and in hindsight. Every cycle has its own character: technology led both the late-1990s expansion and the 2020–2021 recovery, breaking the "early-cycle cyclicals" rule twice; energy led the 2022 slowdown but underperformed in most previous ones. And the phase itself is only knowable with a lag. Use the playbook as a prior, anchor it to this cycle's specific drivers, and size the bets with humility.

¹ Sector playbooks are empirical averages across many cycles, not forecasts. Use them as a prior, then overlay the specific drivers of the cycle you are actually in.

Source: Risk-asset pattern per CFA Institute, Curriculum Vol 2, LM 2, "Market Conditions and Investor Behavior", pp. 55–56. The sector grid is a practitioner overlay.

Deck 4 — Microeconomics & Business Cycles

Own-price elasticity $E_d = (\% \Delta Q) / (\% \Delta P)$; $|E_d| > 1$ elastic, < 1 inelastic. Cut price when elastic to raise revenue; raise price when inelastic. TR peaks where $MR = 0$.

Breakeven at $P = \min ATC$; short-run shutdown at $P < \min AVC$; long-run exit whenever $P < ATC$. Sunk fixed costs are irrelevant to the short-run choice.

One demand curve ($Q = 450 - P$, $MC = 30$) gives three prices: competition 30, Cournot duopoly 170, monopoly 240. More firms push the outcome back toward 30.

CR_N sums shares; HHI sums squares. The five-firm example goes $0.225 \rightarrow 0.410$ on a merger that moves CR_3 by only 10 pp — HHI is the merger-sensitive gauge.

Every market structure maximises profit at $MR = MC$ with MC not falling. Under perfect competition $P = MR$; under every imperfect structure $MR < P$.

Five determinants separate the four structures: sellers, differentiation, barriers, pricing power, non-price competition. Long-run economic profit is zero on the left, positive only where barriers hold.

ArcCo/BatCo Nash = (ArcCo High, BatCo Low) = 650, because Low dominates for BatCo and High dominates for ArcCo. Cooperation pays 800 but needs a side-payment of at least 51.

Four phases set by the output gap. Credit cycles are longer and deeper and lead. Inventory/sales falls, is stable, RISES in slowdown, then normalises. Leading predicts, coincident describes, lagging confirms.